Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following 0.10 m aqueous solutions will have the lowest freezing point?

(A) $\mathrm{Al_2\left(SO_4\right)_3}$

(B) $\mathrm{C_6H_{12}O_6}$

(C) KCl

(D) $\mathrm{C_{12}H_{22}O_{11}}$

Q.2

If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water $\left(\Delta T_f\right)$, when 0.01 mol of sodium sulphate is dissolved in 1 kg of water, is $\left(K_f=1.86\ \mathrm{Kkgmol^{-1}}\right)$

(A) 0.372 K

(B) 0.0558 K

(C) 0.0744 K

(D) 0.0186 K

Q.3

Which one of the following salts will have the same value of van't Hoff factor (i) as that of $\mathrm{K_4\left[Fe(CN)_6\right]}$.

(A) $\mathrm{Al_2\left(SO_4\right)_3}$

(B) NaCl

(C) $\mathrm{Al\left(NO_3\right)_3}$

(D) $\mathrm{Na_2SO_4}$.

Q.4

Coolent used in car radiator is aqueous solution of ethylene glycol. In order to prevent the solution from freezing at $-0.3^{\circ}\mathrm{C}$. How much ethylene glycol (in g) must be added to 5 kg of water ? $\left(\mathrm{K_f=1.86\ K\ kg\ mol^{-1}}\right)$

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(A) 50

(B) 52

(C) 62

(D) 60

Q.5

What is the degree of dissociation of sodium chloride, if the molar mass determined by a cryoscopic method was found to be $31.80\ \mathrm{g\ mol^{-1}}$ [Atomic mass $\mathrm{Na} = 23\ \mathrm{g\ mol^{-1}} \mathrm{Cl} = 35.5\ \mathrm{g\ mol^{-1}}$]?

(A) 0.85

(B) 0.83

(C) 0.84

(D) 0.82

Q.6

A $0.0020\ \mathrm{m}$ aqueous solution of an ionic compound $\mathrm{Co\left(NH_3\right)_5\left(NO_2\right)Cl}$ freezes at $-0.00732^{\circ}\mathrm{C}$. Number of moles of ions which $1\ \mathrm{mol}$ of ionic compound produces on being dissolved in water will be $\left(\mathrm{K_f} = -1.86^{\circ}\mathrm{C/m}\right)$

(A) 2

(B) 1

(C) 3

(D) 4

Q.7

A solution of urea (mol. mass $56\ \mathrm{g\ mol^{-1}}$) boils at $100.18^{\circ}\mathrm{C}$ at the atmospheric pressure. If $\mathrm{K_f}$ and $\mathrm{K_b}$ for water are 1.86 and $0.512\ \mathrm{K\ kg\ mol^{-1}}$ respectively, calculate the temperature in $^{\circ}\mathrm{C}$ at which the above solution will freeze?

(A) -0.654

(B) -0.645

(C) -0.546

(D) -0.564

Q.8

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Freezing point of an aqueous solution is $-0.186^{\circ}\mathrm{C}$. If
the values of K_b and K_f of water are respectively $0.52\ \mathrm{K\ kg\ mol^{-1}}$ and $1.86\ \mathrm{K\ kg\ mol^{-1}}$, then the elevation of boiling
point of the solution in K is

(A) 0.052

(B) 0.050

(C) 0.051

(D) 0.053

Q.9

For preparing $0.1\ \mathrm{N}$ solution of a compound from its impure sample (percentage purity of which is known) weight of the substance required will be

(A) less than the theoretical weight.

(B) more than the theoretical weight.

(C) same as the theoretical weight.

(D) None of these.

Q.10

Which will form maximum boiling point azeotrope?

(A) $\mathrm{HNO_3 + H_2O}$ solution

(B) $\mathrm{C_2H_5OH + H_2O}$ solution

(C) $\mathrm{C_6H_6 + C_6H_5CH_3}$ solution

(D) None of these

Q.11

Lowering of vapour pressure of $1.00\ \mathrm{m}$ solution of a non-volatile solute in a hypothetical solvent of molar mass $40\ \mathrm{g}$ at its normal boiling point, is :

(A) 29.23 torr

(B) 30.4 torr

(C) 35.00 torr

(D) 40.00 torr

Q.12

Equal masses of a solute are dissolved in equal amount of two solvents A and B, respective molecular masses

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being M_A and M_B. The relative lowering of vapour pressure of solution in solvent A is twice that of the solution in solvent B. If the solutions are dilute, M_A and M_B are related as

(A) $M_A = M_B$

(B) $2M_A = M_B$

(C) $M_A = 2M_B$

(D) $M_A = 4M_B$

Q.13

At 20°C and 1.00 atm partial pressure of H_2, $18\ \text{mL}$ of H_2 (STP) dissolves in 1 L of water. If $2\ \text{L}$ of water is exposed to a gaseous mixture having a total pressure of 1425 torr (excluding the vapour pressure of water) and containing $80\%\text{H}_2$ by volume, the volume of H_2 (STP) dissolved is

(A) $27\ \text{mL}$

(B) $54\ \text{mL}$

(C) $33.75\ \text{mL}$

(D) $67.50\ \text{mL}$

Q.14

$12\ \text{g}$ of urea is dissolved in 1 litre of water and $68.4\ \text{g}$ of sucrose is dissolved in 1 litre of water. The lowering of vapour pressure of first case is

(A) equal to second

(B) greater than second

(C) less than second

(D) double that of second

Q.15

How many grams of methyl alcohol should be added to 10 litre tank of water to prevent its freezing at $268\ \text{K}$? (K_f for water is $1.86\ \text{K kg mol}^{-1}$)

(A) $880.07\ \text{g}$

(B) $899.04\ \text{g}$

(C) $886.02\ \text{g}$

(D) $868.06\ \text{g}$

Q.16

Which one of the following aqueous solutions will exihibit highest boiling point?

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(A) $0.015\mathrm{M}$ urea

(B) $0.01\mathrm{MKNO_3}$

(C) $0.01\mathrm{MNa_2SO_4}$

(D) $0.015\mathrm{M}$glucose

Q.17

The total vapour pressure of a 4 mole $\%$ solution of $\mathrm{NH_3}$ in water at $293\ \mathrm{K}$ is 50.0 torr. The vapour pressure of pure water is 17.0 torr at this temperature. Applying Henry's and Raoult's laws, the total vapour pressure for a 5 mole $\%$ solution is

(A) 58.25 torr

(B) 33 torr

(C) 42.1 torr

(D) 52.25 torr

Q.18

KBr is 80% ionized in solution. The freezing point of 0.4 molal solution of KBr is :
$\mathrm{K_f\ (H_2O) = 1.86\ K\ kg/mole}$

(A) $274.339\ \mathrm{K}$

(B) $-1.339\ \mathrm{K}$

(C) $257.3\ \mathrm{K}$

(D) $-1.339^{\circ}\mathrm{C}$

Q.19

A solution of urea (mol. mass $56\ \mathrm{g\ mol^{-1}}$) boils at $100.18^{\circ}\mathrm{C}$ at the atmospheric pressure. If $\mathrm{K_t}$ and $\mathrm{K_b}$ for water are 1.86 and $0.512\ \mathrm{K\ kg\ mol^{-}}$
respectively, the above solution will freeze at

(A) $0.654^{\circ}\mathrm{C}$

(B) $-0.654^{\circ}\mathrm{C}$

(C) $6.54^{\circ}\mathrm{C}$

(D) $-6.54^{\circ}\mathrm{C}$

Q.20

A 0.120 molal solution of CsCl [with ionizes in the solution as $\mathrm{CsCl \rightleftharpoons Cs^{+} + Cl^{-}}$] freezes at -$0.4^{\circ}\mathrm{C}$. The Van't Hoff factor and degree of dissociation of CsCl in the solution, respectively, are
$\mathrm{(Given\ K_f water = 1.86\ \ K\ \ kg\ \ mol^{-1})}$

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(A) $1.79,\ 0.79$

(B) $1.5,\ 0.5$

(C) $1.92,\ 0.92$

(D) $1.2,\ 0.2$

Answers & Solutions

Q.1 Answer:

$Al_2\left(SO_4\right)_3$

Solution:

Depression in F.P. $\propto$ **No. of particles.**

$Al_2\left(SO_4\right)_3$ **provides five ions on ionisation**

$$Al_2\left(SO_4\right)_3 \longrightarrow 2Al^{3+} + 3SO_4^{2-}$$

while KCl provides two ions

$$KCl \rightleftharpoons K^+ + Cl^-$$

$C_6H_{12}O_6$ **and** $C_{12}H_{22}O_{11}$ **are not ionised so they have single particle. Hence,** $Al_2\left(SO_4\right)_3$ **have maximum value of depression in F.P or lowest F.P**

Q.2 Answer:

$0.0558\ K$

Solution:

Sodium sulphate dissociates as $Na_2SO_4(s) \longrightarrow 2Na^+ + SO_4^{2-}$

hence van't hoff factor $i = 3$

Now $\Delta T_f = ik_f \cdot m$

$= 3 \times 1.86 \times 0.01 = 0.0558\ K$

Q.3 Answer:

$Al_2\left(SO_4\right)_3$

Solution:

$K_4\left[Fe(CN)_6\right]$ **and** $Al_2\left(SO_4\right)_3$ **both dissociates to give 5 ions or** $i = 5$ $K_4\left[Fe(CN)_6\right] \rightleftharpoons 4\ K^+ + \left[Fe(CN)_6\right]^4$

and $Al_2\left(SO_4\right)_3 \rightleftharpoons 2Al^{3+} + 3SO_4^{2-}$

Q.4 Answer:

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Solution:

$\Delta T_f = 0.3^\circ C$
$\Delta T_f = 0.3^\circ C = \frac{K_f \times W_B \times 1000}{M_B \times W_A}$
$0.3 = \frac{1.86 \times W_B \times 1000}{62 \times 5000}$
$\therefore \quad W_B = 50\ g$
The amount used should be more than $50\ g$

Q.5 Answer:

0.83

Solution:

Let α **be the give of dissociati** n **then**
Van't Hoff's factor $i = \frac{1-\alpha+\alpha+\alpha}{1} = 1+\alpha$
Again Van't Hoff's factor
$= \frac{\text{Normal mol. wt}}{\text{Observed mol. wt}} = \frac{58.5}{31.8} = 1.83$
Equating to both values of i, $\quad \therefore 1+\alpha = 1.83$
$\therefore \alpha = 0.83$

Q.6 Answer:

2

Solution:

$\Delta T_f = 0 - (-0.00732^\circ) = 0.00732$
$\Delta T_f = i \times K_f \times m$
$i = \frac{\Delta T_f}{K_f \times m} = \frac{0.00732}{1.86 \times 0.002} = 2$

Q.7 Answer:

-0.654

Solution:

$(\mathbf{-0.654})$ **As** $\Delta T_f = K_f m$
$\Delta T_b = K_b.m$
Hence, we have $m = \frac{\Delta T_f}{K_f} = \frac{\Delta T_b}{K_b}$
or $\Delta T_f = \Delta T_b \frac{K_f}{K_b}$
$\Rightarrow [\Delta T_b = 100.18 - 100 = 0.18^\circ C]$
$= 0.18 \times \frac{1.86}{0.512} = 0.654^\circ C$
As the Freezing Point of pure water is $0^\circ C$, $\Delta T_f = 0 - T_f$
$0.654 = 0 - T_f$
$\therefore T_f = -0.654$
Thus the freezing point of solution will be $-0.654^\circ C$.

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Q.8 Answer:

0.052

Solution:

$\Delta T_f = i.K_f \cdot m; \Delta T_b = i.K_b \cdot m$
$\frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b}$
$\Delta T_f = 0 - (-0.186^\circ\text{C}) = 0.186^\circ\text{C}$
$\frac{0.186}{\Delta T_b} = \frac{1.86}{0.52} \Rightarrow \Delta T_b = \frac{0.52 \times 0.186}{1.86} = 0.052$

Q.9 Answer:

more than the theoretical weight.

Solution:

More than theoretical weight since impurity will not contribute.

Q.10 Answer:

$HNO_3 + H_2O$ solution

Solution:

The solutions (liquid mixture) which boils at constant temperature and can distil as such without any change in composition are called azeotropes. Solution of HNO_3 and H_2O will form maximum boiling point azeotrope. Maximum boiling azeotropes show negative deviation from Raoult's law.
Composition (\%)Boiling Point HNO_3
$68.0 \quad 359\ K$
H_2O
32.0
$373\ K$
Boiling point of the azeotrope of these two solutions is $393.5\ K$.

Q.11 Answer:

29.23 torr

Solution:

$P^0_{\text{solvent}} = 760$ **torr (at the b. pt.)** $n_{\text{solute}} = 1$
$n_{\text{solvent}} = \frac{1000}{40} = 25$
$\Delta P = 760 \times X_{\text{solute}} = \frac{760}{26}$ **torr** $= 29.23$ **torr**

Q.12 Answer:

$M_A = 2M_B$

Solution:

For dilute solution,
$\frac{\Delta P}{P^\circ} \Rightarrow \frac{n_{\text{solute}}}{n_{\text{solvent}}}$ **For solution in A,**

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$\frac{\Delta P_A}{P_A^\circ} = \frac{W/M}{W_A/M_A} = \frac{W}{M} \times \frac{M_A}{W_A}$

For solution in B, $\frac{\Delta P_B}{P_A^\circ} = \frac{W}{M} \times \frac{M_B}{W_B}$ **. . ...(ii)**

From (i) and (ii), $\frac{\Delta P_A/P_A^\circ}{\Delta P_B/P_B^\circ} = 2 = \frac{M_A W_B}{M_B W_A} = \frac{M_A}{M_B}\ (W_A = W_B)$

$M_A = 2M_B$

Q.13 Answer:

$54\ \text{mL}$

Solution:

$V_{\text{dissolved}} = KP_{H2}$

P_{H2} **in the mixture** $= X_{H2}P = \frac{80 \times 1425}{100} = 1140$

torr $= \frac{1140}{760} = 1.5\ \text{atm}$

Hence, $V_{\text{dissolved}}$ **in 2 L of water** $= 2 \times K \times P_{H2}$

$= 2 \times 18 \times 1.5 = 54\ \text{mL}$

Q.14 Answer:

equal to second

Solution:

Moles of urea $= \frac{12}{60} = 0.2$

Moles of sucrose $= \frac{68.4}{342} = 0.2$

Both are non electrolyte hence lowering of V.P. will be same.

Q.15 Answer:

$868.06\ \text{g}$

Solution:

$\Delta T_f = K_f m$

where m **= molality**

$273 - 268 = 1.86 \times \frac{w}{M \times v}$

$5 = 1.86 \times \frac{w}{32 \times 10}$

$w = \frac{5 \times 32 \times 10}{1.86}$

$= 860.2 \approx 868.06\ \text{g}$

Q.16 Answer:

$0.01M Na_2SO_4$

Solution:

$\Delta T_b = K_b \times m$

Elevation in boiling point is a colligative property, which depends upon the no. of particles (concentration of solution). Thus greater the number of particles, greater is the elevation in boiling

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point and hence greater will be its boiling point.

$$Na_2SO_4 \rightleftharpoons 2Na^+ + SO_4^{2-}$$

Since Na_2SO_4 has maximum number of particles hence it has maximum boiling point.

Q.17 Answer:

58.25 torr

Solution:

The given data are $P_{water} = 17.0$ **torr**

P_{total} (4 **mole** $\%$ **solution** $) = P_{NH_3} + P_{water} = 50.0$ **torr**

$X_{NH_3} = 0.04$ **and** $X_{water} = 0.96$

Now according to Raoult's law;

$P_{water} = X_{water} P^\circ_{water}$

$= 0.96 \times 17.0$ **torr** $= 16.32$ **torr**

Now Henry's law constant for ammonia is

$K_H (NH_3) = \frac{P_{NH_3}}{X_{NH_3}} = \frac{33.68 \text{ torr}}{0.04} = 842$ **torr**

Hence, for 5 mole $\%$ **solution, we have**

$P_{NH_3} = K_H (NH_3) X_{NH_3}$

$= (842$ **torr** $)(0.05) = 16.15$ **torr**

Thus, P_{total} (5 **mole** $\%$ **solution)**

$= P_{NH_3} + P_{water} = 42.1 + 16.15 = 58.25$ **torr**

Q.18 Answer:

$-1.339^\circ C$

Solution:

$\Delta T_f = iK_f m$

$KBr \longrightarrow K^+ \quad + \quad Br$

1

0

$(1-\alpha) \quad \alpha$

$i = 1 + \alpha$

Here $\alpha = 0.8$ **(80\% ionization)** $= 1.8$

$\Delta T_f = (1.8)(1.86)(0.4) = 1.339$

$T_f = -1.339^\circ C$

Q.19 Answer:

$-0.654^\circ C$

Solution:

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As $\Delta T_f = K_f m$

$\Delta T_b = K_b \cdot m$

Hence, we have $m = \frac{\Delta T_f}{K_f} = \frac{\Delta T_b}{K_b}$

or $\Delta T_f = \Delta T_b \frac{K_f}{K_b}$

$\Rightarrow [\Delta T_b = 100.18 - 100 = 0.18^\circ C]$

$= 0.18 \times \frac{1.86}{0.512} = 0.654^\circ C$

As the freezing point of pure water is $0^\circ C$ $\Delta T_f = 0 - T_f$

$0.654 = 0 - T_f$

$\therefore T_f = -0.654$

Thus the freezing point of solution will be $-0.654^\circ C$

Q.20 Answer:

1.79, 0.79

Solution:

$\Delta T_f = i \times k_f \times m$

$T_f^\circ - T_f = 1.86 \times i \times 0.120$

$0.4 = i \times 0.2232$

$i = 1.79$

$i = 1 + (n - 1)\alpha$ $n = 2$

$1.79 = 1 + \alpha$

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